Artificial Intelligence Lab

Lab Report # 04

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Implementation of OOP Concepts in Python

Objectives: -

- To understand basics of Object and Classes.
- Implementing Constructors and its types.
- Implementing inheritance.

Question # 01

Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

Solution: -

```python
import math
class Circle:  1 usage
    def __init__(self, radius):
        self.radius = radius
    def area(self):  1 usage
        return math.pi*self.radius ** 2

    def perimeter(self):  1 usage
        return 2*math.pi*self.radius
c = Circle(4)
print("Area:", c.area())
print("Perimeter:", c.perimeter())
```

Output: -

```
Area: 50.26548245743669
Perimeter: 25.132741228718345

Process finished with exit code 0
```

Question # 02

Write a Python class to implement pow (x, n). Do not use built-in math.pow () function for this code. Also handle this code for all +ve as well as -ve numbers.

Solution: -

```python
def custom_pow(x,n):  2 usages
    if n==0:
        return 1
    elif n>0:
        result=1
        for _ in range(n):
            result*= x
        return result
    else:
        result=1
        for _ in range(-n):
            result *= x
        return 1/result
print(custom_pow( x: 2,  n: 3))
print(custom_pow( x: 2, -2))
```

Output: -

```
8
0.25

Process finished with exit code 0
```

Question # 03

Write a Python code for the following figure.

Solution: -

```python
class Item:
    def __init__(self,name,price):
        self.name=name
        self.price=price
    def view(self):
        print(f"{self.name}-Rs.{self.price}")
class MP3(Item):
    def play(self):
        print(f"Playing {self.name}")
class DVD(Item):
    def trailer(self):
        print(f"Trailer of {self.name}")
class Book(Item):
    def preview(self):
        print(f"Previewing {self.name}")

mp3=MP3( name: "NFAK",  price: 150)
dvd=DVD( name: "SALT",  price: 500)
book=Book( name: "Forty Rules of Love ",  price: 700)

mp3.view()
mp3.play()

dvd.view()
dvd.trailer()

book.view()
book.preview()
```

Output: -

```
NFAK - Rs.150
Playing NFAK
SALT - Rs.500
Trailer of SALT
Forty Rules of Love  - Rs.700
Previewing Forty Rules of Love

Process finished with exit code 0
```

Task:

Employee Management:

- Create a Person class with attributes/variables “name” and “age”.
- Create a Job class with attributes/variables “designation” and “salary”.
- Create an Employee class that inherent the above both classes
- Implement a method to display employee details.

Solution: -

```python
class Person:  2 usages
    def __init__(self, name, age):
        self.name = name
        self.age = age
class Job:  2 usages
    def __init__(self, designation, salary):
        self.designation = designation
        self.salary = salary
class Employee(Person, Job):  1 usage
    def __init__(self, name, age, designation, salary):
        # Initialize Person attributes
        Person.__init__(self, name, age)
        # Initialize Job attributes
        Job.__init__(self, designation, salary)
    def display_details(self):  1 usage
        print(f"Employee Name: {self.name}")
        print(f"Age: {self.age}")
        print(f"Designation: {self.designation}")
        print(f"Salary: {self.salary}")
emp1 = Employee( name: "Shaheer Ahmad",  age: 23,  designation: "Cyber Security Analyst",  salary: 120000)
emp1.display_details()
```

Output: -

```
Employee Name: Shaheer Ahmad
Age: 23
Designation: Cyber Security Analyst
Salary: 120000

Process finished with exit code 0
```